

For each of the following functions,

1. calculate the gradient,
2. calculate the Hessian,
3. check if the function is convex, concave or neither,
4. calculate the curvature of the function in the direction d at the specified point x^+ .

Hint: remember that a symmetric matrix is positive definite if all its eigenvalues are positive.

$$\begin{aligned}f(x) &= \frac{1}{2}x_1^2 + \frac{9}{2}x_2^2, \\x^+ &= (0, 0)^T, \\d &= (1, 1)^T.\end{aligned}$$

$$\begin{aligned}g(x) &= \frac{1}{3}x_1^3 + x_2^3 - x_1 - x_2, \\x^+ &= (9, 1)^T, \\d &= (9, 1)^T.\end{aligned}$$